Scalar-Carrier Smooth-Inhomogeneity Current-Closure Release for HAOS-IIP

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Abstract

This paper freezes release `52.5` of HAOS-IIP as the bounded follow-on to the clean-line shell-native current closure `52.4`. Release `52.4` showed that, on the undeformed scalar carrier, the shell-native transient reconstruction closes onto one bounded constitutive family. Release `52.5` asks the next stricter question without changing the carrier: if the point cloud is deformed in a controlled way, do the extracted local metric field and the shell-native current law co-deform together? The answer is mixed but positive on the smooth branch. All six smooth radial deformation cases pass across amplitudes $\eta = 0.05, 0.10, 0.15$ and clean refinements $n = 13, 15$, with fitted conductivity staying close to the heat diffusivity, bounded residuals, high metric-tracking correlation, and bounded refinement drift. By contrast, a localized bump deformation remains open at the same amplitudes. The correct conclusion is therefore narrow: on the same scalar carrier, the local metric read and the shell-native transient constitutive law co-deform together under smooth radial inhomogeneity, while more localized inhomogeneity remains outside the same bounded closure. This is a controlled smooth-inhomogeneity transport result, not yet disorder universality, conserved-current closure, curvature extraction, or spacetime physics.

Contents

1 Scope

Release `52.5` is a bounded continuation of the scalar transport line.

It does not add:

- arbitrary inhomogeneity closure;
- disorder-universal transport;
- a conserved-current theorem;
- curvature extraction;
- spacetime or ontology language.

It does add:

- a controlled inhomogeneity test on the same scalar carrier;
- a bounded smooth radial co-deformation statement linking metric extraction and shell-native transport closure;
- an explicit boundary showing that localized bump inhomogeneity remains open.

2 Where 52.4 Left the Boundary

Release `52.4` already closed the clean refinement line:

- same local scalar kernel graph;
- same coarse local metric field from `51.4`;
- same shell-native transient constitutive law.

Its strongest bounded conclusion was:

> on the clean scalar carrier, a shell-native transient current reconstruction closes onto one bounded constitutive family whose fitted conductivity stays close to the heat diffusivity and whose shellwise profile remains refinement-stable.

The next honest question was therefore not whether the clean line closes, but whether that closure survives a controlled deformation of the same carrier.

3 Controlled Inhomogeneity Design

Release `52.5` keeps the carrier and reconstruction fixed and changes only the point cloud through deterministic deformation profiles.

Two families are tested:

- smooth radial deformation;
- localized scalar bump.

The amplitudes are:

$$\eta \in \{0.05,\ 0.10,\ 0.15\},$$

and the clean refinements are:

$$n \in \{13,\ 15\}.$$

For each case, the repository:

- deforms the clean scalar carrier;
- rebuilds the scalar operator on the deformed point cloud;
- extracts the coarse shellwise metric field;
- fits the shell-native constitutive current law against transient shell flow;
- checks whether the metric shift and the constitutive profile track the imposed deformation in a bounded way.

4 Frozen Quantitative Summary

4.1 Smooth radial branch

Case	κ/D_{eff}	median err.	$p90$ err.	shell CV	$\lvert\text{corr}\rvert$	status
`n=13`, $\eta = 0.05$	0.9019	0.0704	0.3240	0.1193	0.9885	PASS
`n=13`, $\eta = 0.10$	0.9243	0.0665	0.3430	0.1265	0.9939	PASS
`n=13`, $\eta = 0.15$	0.9466	0.0568	0.3568	0.1310	0.9840	PASS
`n=15`, $\eta = 0.05$	0.9272	0.0539	0.2429	0.1072	0.9872	PASS
`n=15`, $\eta = 0.10$	0.9588	0.0496	0.2686	0.1178	0.9935	PASS
`n=15`, $\eta = 0.15$	0.9848	0.0562	0.2773	0.1215	0.9846	PASS

Table 1: All smooth radial inhomogeneity cases pass the bounded `52.5` criteria.

The radial refinement-drift read also stays bounded:

- $\eta = 0.05$: drift $= 0.1202$;
- $\eta = 0.10$: drift $= 0.1384$;
- $\eta = 0.15$: drift $= 0.1465$.

Case	κ/D_{eff}	median err.	$p90$ err.	shell CV	$\lvert\text{corr}\rvert$	status
`n=13`, $\eta = 0.05$	0.7857	0.2053	0.4658	0.2164	0.9752	OPEN
`n=13`, $\eta = 0.10$	0.6876	0.3113	0.6112	0.3331	0.9743	OPEN
`n=13`, $\eta = 0.15$	0.5889	0.4299	0.7346	0.4963	0.9628	OPEN
`n=15`, $\eta = 0.05$	0.7971	0.1569	0.3896	0.1944	0.9924	OPEN
`n=15`, $\eta = 0.10$	0.6929	0.2487	0.5368	0.3088	0.9909	OPEN
`n=15`, $\eta = 0.15$	0.5929	0.3565	0.6609	0.4453	0.9853	OPEN

Table 2: Localized bump inhomogeneity does not satisfy the same bounded closure criteria.

4.2 Localized bump branch

The bump refinement drift is also outside the smooth branch window:

- $\eta = 0.05$: drift $= 0.1922$;
- $\eta = 0.10$: drift $= 0.3150$;
- $\eta = 0.15$: drift $= 0.5124$.

5 What 52.5 Now Supports

The strongest honest statement now supported by the repository is:

> on the same scalar carrier, the extracted local metric field and the shell-native transient constitutive law co-deform together under smooth radial inhomogeneity, while more localized inhomogeneity remains outside the same bounded closure.

This extends the scalar line from:

- local metric closure;
- shell-native inverse-square response closure;
- clean-line shell-native transient current closure;

to:

- bounded smooth-inhomogeneity transport closure on the same carrier.

That is more physics-facing than `52.4` alone because the transport closure is no longer restricted to the undeformed clean line.

6 What 52.5 Still Does Not Say

Release `52.5` still does not justify:

- arbitrary or sharply localized inhomogeneity closure;
- disorder-universal current laws;
- a conserved-current theorem;
- curvature extraction;
- spacetime or ontology language.

So `52.4` remains the clean-line closure, and `52.5` is the bounded smooth-inhomogeneity follow-on.

7 Authority and Artifacts

The release is backed by:

- operator note: `experiments/operators/Scalar_Kernel_Graph_Current_Closure_Inhomogeneity_v1.md`;
- script: `numerics/simulations/scalar_kernel_graph_current_closure_inhomogeneity.py`;
- config key: `config.json -> scalar_kernel_graph_current_closure_inhomogeneity`;
- frozen result: `data/20260422_213513_scalar_kernel_graph_current_closure_inhomogeneity.json`;
- latest result: `data/scalar_kernel_graph_current_closure_inhomogeneity_latest.json`.

8 The Right Next Move

The next scalar step should stay narrow:

- test whether the smooth radial branch survives mild carrier disorder;
- measure the localization threshold where the bump branch stops behaving like the smooth branch;
- only afterward revisit conserved-current or curvature-style closure.

9 Conclusion

Release `52.5` freezes the first controlled inhomogeneity result on the scalar transport line.

It does not say that inhomogeneous transport is solved in general. It says something narrower and more useful: on the same scalar carrier, the local metric read and the shell-native transient constitutive law co-deform together under smooth radial inhomogeneity, while localized bump deformations remain explicitly open. That is the correct stopping point for `52.5`: stronger than a clean-line closure alone, still disciplined about the boundary that has not yet been crossed.